Summary Report: Startup Viability Analysis

Project Summary

This project investigates the market potential for a portable, AI-powered gardening robot designed to encourage sustainable gardening practices. The goal is to identify and understand the needs of potential customers to determine if it is viable to create and launch this product. If so, the insights gained will guide further product design and marketing strategy.

This data analysis project is in progress, and the initial keyword analysis stage is completed. Relevant keywords with high audience interest have been identified, and potential customer segments have been explored. Further keyword refinement is ongoing. The final selection of keywords is underway, followed by their formatting for efficient data collection.

After keyword selection, the finalized keywords will be used to collect data from online sources. Feature engineering techniques will be employed to analyze the data, identifying common traits and behaviors among potential customers, leading to the creation of initial customer profiles or groups. These groups will be further refined into distinct market segments by applying a clustering algorithm. Customer segments will then be used to estimate the total market size and potential revenue for each segment as well as the overall market, providing essential quantitative data to assess the viability and potential market success of the startup idea.

Methodology

The search trend and audience interest analysis phase involved gathering Google Trends data over the past three years using the unofficial Google Trends Python API. Input consisted of three keyword lists: prototype features, needs addressed, and problems solved. A custom function efficiently retrieved this data, employing the API's "interest over time" method and batch processing. Exploratory Data Analysis (EDA) was conducted using Pandas and Seaborn to visualize and explore the interest-over-time data. Bar plots, line plots, and heatmaps were employed to depict overall interest, weekly averages, yearly and monthly patterns, and comparisons across features, problems, and needs groups. The cleaned and formatted data was then stored in Google Cloud Storage for subsequent analysis.

Next, keywords were categorized into 'Critical' (75-100%), 'High' (50-75%), 'Moderate' (10-50%), and 'Low' (<10%) relevance groups based on their interest compared to the highest interest keyword within each group. Summary statistics were calculated for 'Critical,' 'High,' and 'Moderate' keywords using Pandas' `describe` method. Further analysis involved statistical modeling using SciPy's `linregress` and `find_peaks` methods. Linear regression analysis revealed the direction and magnitude of current interest trends, while peak detection provided insights into overall interest magnitude, trends over time, and interest stability. Finally, a qualitative assessment of Google Search suggestions was performed to gain deeper audience insights.

The analysis leveraged Python as the programming language, utilizing Pandas for data manipulation and Seaborn and Matplotlib for visualizations. HoloViews with the Bokeh extension were employed for interactive plotting. SciPy facilitated statistical modeling, and Google Cloud Storage was used for data storage and management. All tasks were conducted within Google Colab notebooks, promoting a streamlined workflow that encouraged experimentation and comprehensive documentation.

Results & Insights

This section summarizes current findings and insights. It will be updated as the project evolves to provide a more comprehensive understanding of the target audience and market opportunity. Additional visualizations and the underlying code and data are available in the project's GitHub repository [1].

Key Findings from the Initial Keyword Analysis

- The analysis identified several promising keywords with high interest or significant growth potential, including features like ”solar-powered,” “plant detection,” and “remote access and control,” and needs like ”educational AI,” “climate resilience,” and “sustainable food production.” These keywords have a significant audience and can be used for data collection in the subsequent analysis step.

- Potential audience segments were identified: solar consumers, academic innovators, climate activists, health-conscious consumers, and forward-thinking farmers. These segments provide an initial understanding of the diverse groups that might be interested in the gardening robot prototype.

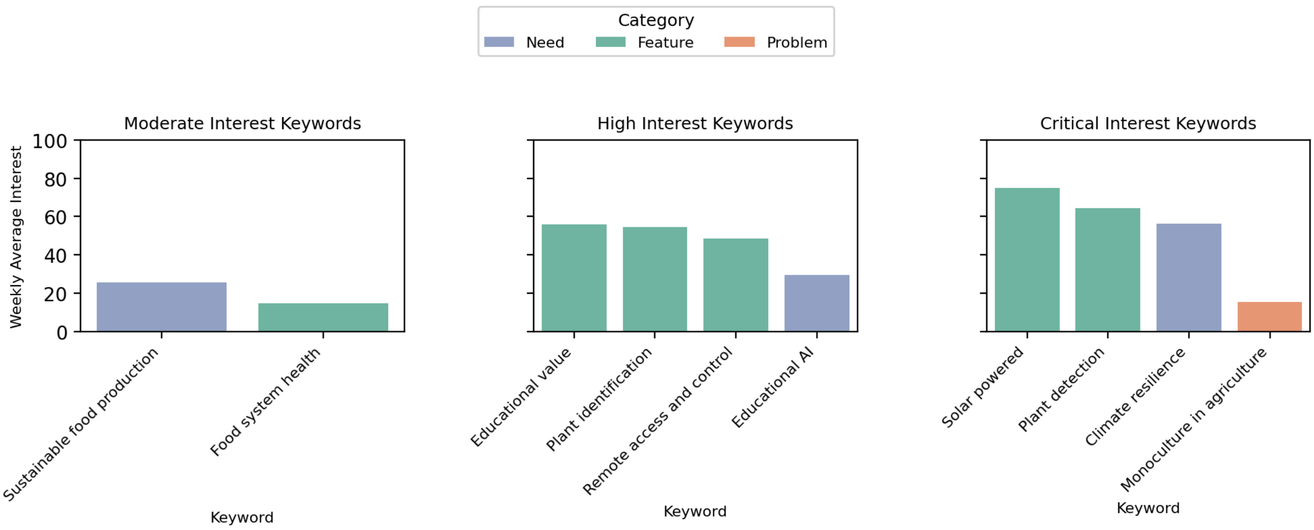
- The broader interest categories associated with these audiences include science and technology, food and agriculture, and energy and environment. These categories offer potential areas for further exploration and targeting and can be used in conjunction with keywords for data collection.

- Some keywords require further evaluation before determining their suitability for inclusion in subsequent analysis stages. These include the features “food system health” and “plant identification” and the entire problems keyword list, which needs refinement. In the problems group, the only keyword that showed some potential was “monoculture in agriculture.”

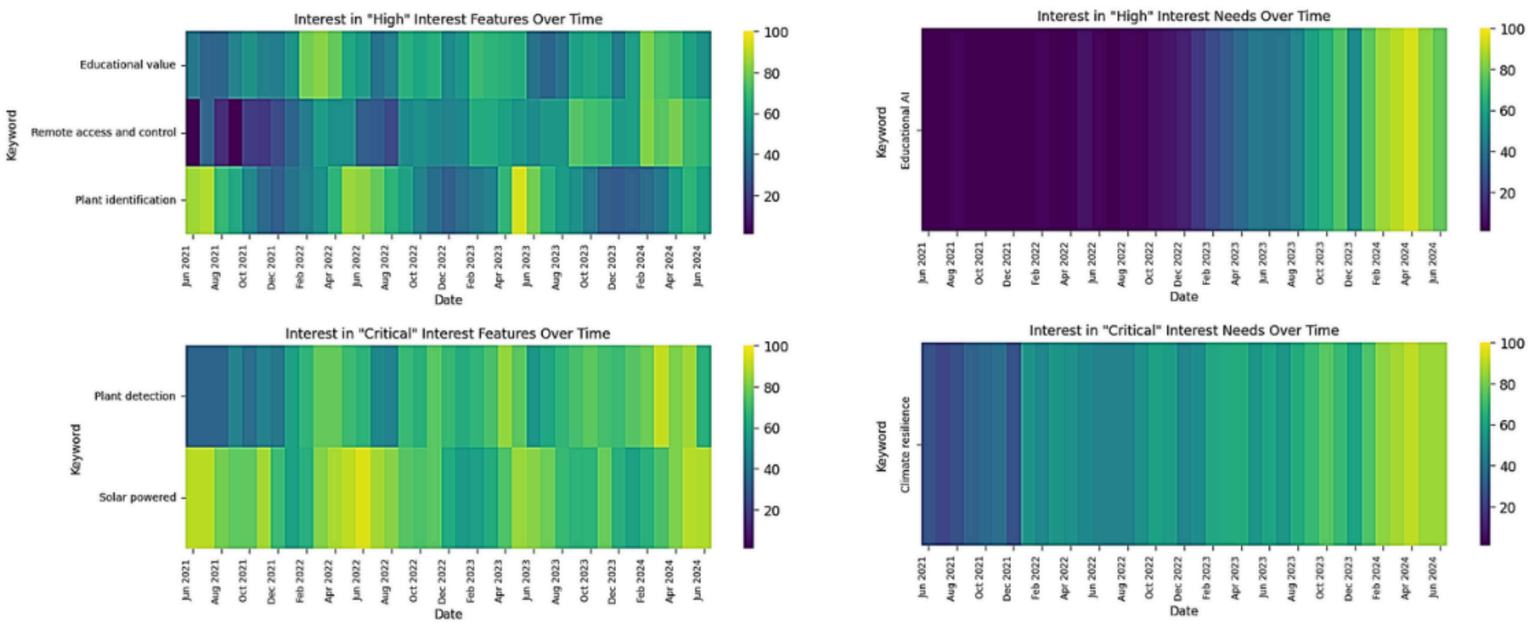
Interpretation of Results & Insights

Key findings from the keyword analysis include identifying promising keywords and potential customer segments. The initial analysis narrowed the focus to the most relevant and promising areas for further investigation and eliminated keywords of little interest. The insights gained enable subsequent analysis steps to focus exclusively on keywords with significant market potential.

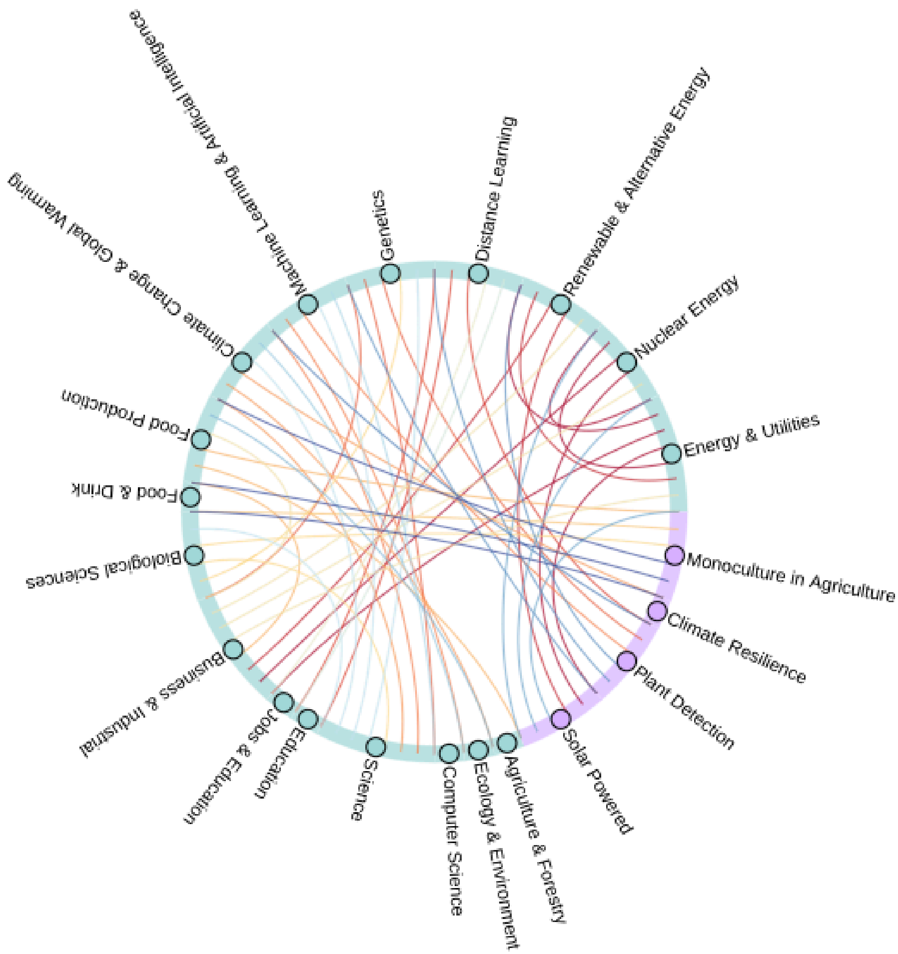
Visualizations



Above: This bar chart shows the weekly average interest for the most relevant keywords identified in the Google Trends analysis. It is categorized by type (need, feature, or problem) and interest level (moderate, high, or critical). It shows absolute interest in keywords in all three keyword groups. Interest is quantified on a scale from 1 to 100, where 100 represents peak interest relative to interest in all other keywords globally, and 0 indicates little to no global interest. The weekly average interest was calculated by aggregating data over the last three years. Keywords in the features group received the most interest.



Above: These four heatmaps display the Google Trends interest over time for the most relevant features and needs, categorized by interest level (high, critical). Keywords in the features group are displayed on the left side, and those in the needs group are displayed on the right. Features shown (top left to bottom right) are "Educational value," "Remote access and control," "Plant identification," "Plant detection," "Solar powered," "Educational AI," and "Climate resilience." These heatmaps show growth patterns and continuously high interest in the most relevant keywords.



Left: This chord diagram illustrates the connections between the identified keywords and their associated Google Search categories, showcasing the interconnectedness of the topics relevant to the target audience. This analysis used only keywords classified as of *critical* interest in their groups.

Next Steps

The following steps will focus on refining keyword selection, particularly in the "problems" domain, and formatting the relevant keywords for use as input into APIs to collect web and social media data. Subsequent analysis will utilize this gathered data to identify and characterize distinct audience segments interested in the AI gardening robot, informing prototype design and market opportunity assessment through customer segmentation and TAM estimation.

Appendix

1) For further details and access to the underlying code and data, please refer to the project's GitHub repository at: https://github.com/alexhosp/startup-viability-analysis/tree/main